INCH-POUND

MIL-O-70548 (AR)
AMENDMENT 6
14 May 1997
SUPERSEDING
AMENDMENT 5
05 April 1995

MILITARY SPECIFICATION

OBTURATING BAND, PLASTIC SPLIT (MOLDED)

This amendment forms a part of MIL-O-70548 (AR), dated 5 September 1986, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, Change "M853" To "M853A1".

PAGE 2

* 2.2, Under American Society Of Testing And Material (ASTM):
Add:

"ASTM-D1238 - Test for Flow Rates of Thermoplastics by Extrusion Plastometer."

Delete:

"ASTM-D648 - Test for Deflection Temperature of Plastics under Flexural Load."

AMSC N/A **FSC 1395**

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Delete:

"ASTM-D3935-80 - Polycarbonate (PC) Unfilled and Reinforced Materials."

and substitute:

"ASTM-D3935-87 - Polycarbonate (PC) Unfilled and Reinforced Material".

* 3.2: Delete in its entirety and substitute:

"3.2 Molded parts. Molding pellets, molded obturators and meterial test specimens shall comply with the applicable requirements specified on drawing 9310339. The molded obturators shall be produced by the injection molding process."

* 3.4: Delete in its entirety and substitute:

"3.4 Physical property of material. Material used to produce the obturators shall meet the properties listed as follows (ASTM-D3935-87):

CHARACTERISTIC	REQUIREMENTS
Izod impact strength, min.	750 Joules/meter (J/M)
Tensile yield strength, min.	57 Megapascal (Mpa)
Melt flow rate	3.0 - 9.0 gram/10 minutes
Flexural modulus, min.	2000 Megapascal (Mpa)".

Add new paragraphs 3.7 and 3.8 as follows:

"3.7 Proving ground test. The mean muzzle velocity of the test rounds which use the test obturators shall not differ more than 5 fps from that of the comparison samples with accepted obturators when determined by the statistical test (see 4.4.5.1.8). In flight, the obturator shall come off the test round within 20 feet from the muzzle of the mrotar tube (see 4.4.3.2).

3.8 Chemical composition. The chemcial composition for each lot of material shall be analyzed and furnished in accordance with 6.7 to ARDEC, ATTN: AMSTA-AR-QAA-C, Picatinny Arsenal, NJ 07806-5000."

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4.3.3, Delete in its entirety and substitute:

"4.3.3 Rejection. See MIL-A-48078 and 4.4.3.2."

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* TABLE I, First Article Inspection: Delete in its entirety and substitute:

TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

	TITLE Ring, Obturating	SHEE 1 OF 1		DRAWING NUMBER See below
				NEXT HIGHER ASSEMBLY N/A
	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Ring, Obturating (dwg. 9310399)			
	Examination for defects	100	3.2/3.3	4.4.2.1
	Proving Ground test	70	3.7	4.4.3.2
	Physical Properties of Material Test Specimens			
	Izod impact strength (min)	5	3.4	4.4.3.1/4.4.5.1.1
	Tensile yield strength	5	3.4	4.4.3.1/4.4.5.1.2
	Flexural Modulus	5	3.4	4.4.3.1/4.4.5.1.4
	Physical Properties of Molded Pellets			
	Melt flow rate	1 oz.	3.4	4.4.3.1/4.4.5.1.3 1/
	Physical Properties of Molded Obturators			
	Melt flow rate	1 oz.	3.4	4.4.3.1/4.4.5.1.3 1/

NOTES:
1/ Both the molding pellets and the obturators must contain moisture not greater than .020 percent of weight for use in the test.

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4.4.1.1, Delete: "batch or run" and substitute: "lot".

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* 4.4.2.1, Major 105: Delete: "Presence of voids ... 0.40% ... 3.2 ... Visual" and substitute:

"Presence of voids, molding flash or surface splay ... 0.40% ... 3.2 ... Visual"

* Major 107: Delete in its entirety and substitute:

107	Physical Properties of Material Test Specimens			
	Izod impact strength (min)	5	3.4	4.4.3.1/4.4.5.1.1
	Tensile yield strength	5	3.4	4.4.3.1/4.4.5.1.2
	Flexural Modulus	5	3.4	4.4.3.1/4.4.5.1.4
	Physical Properties of Molded Pellets			
	Melt flow rate	1 oz.	3.4	4.4.3.1/4.4.5.1.3 1/
	Physical Properties of Molded Obturators			
	Melt flow rate	1 oz.	3.4	4.4.3.1/4.4.5.1.3 1/

NOTES:
1/ Both the molding pellets and the obturators must contain moisture not greater than .020 percent of weight for use in the test.

* Add Major 108 as follows:

108	Moisture content	3 pellets	3.2	Test/4.4.5.1.2

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4.4.3.1, Delete in its entirety and substitute:

"4.4.3.1 Physical properties of material. Representative specimens prepared from each lot of plastic material utilized for fabricating the first article and production lots of obturating rings shall be tested in accordance with 4.4.5.1. Five specimen samples shall be subjected to each physical property test. If any sample fails to meet any of the requirements specified in Section 3.4, the lot of the plastic material shall be rejected."

* Add new paragraph 4.4.3.1.a, 4.4.3.1.b, and 4.4.3.1.c as follows:

"4.4.3.1.a Melt flow rate of molding pellet. The contractor shall select approximately one (1) ounce of molding pellet samples to perform the test from each eight (8) hours of production. The melt flow rate of the molding pellets shall comply with the drawing requirement or the eight hours production represented by the samples shall be rejected. The melt flow rate data will be compared to that of the molded obturators.

4.4.3.1.b Melt flow rate of molded obturator. The contractor shall select five (5) molded obturator samples from each eight (8) hours of production. The samples shall be cut into small pieces for this test. If the melt flow rate of the molded obturators is more than that of the molding pellets by 60 percent, the eight hours of production represented by the samples shall be rejected.

4.4.3.1.c Moisture content. The contractor shall provide adequate controls to assure compliance with the requirements and shall select a minimum of three (3) molding pellets at the begining of each eight (8) hours of production using the General Electric Tomasetti Volatile Indicator (TVI) moisture test or a properly calibrated moisture analyzer that is approved by ARDEC."

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Add new paragraph 4.4.3.2 as follows:

"4.4.3.2 Proving ground test. Forty-five test obturators and twenty-five comparison obturators shall be selected for this test (see Table II). The test shall be performed as specified in 4.4.5.1.6. The lot shall be rejected if one or more groups of the test obturator fails to meet the accept/reject criteria in Table III.

Table II. Ballistic Performance.

Group	Obturator	Projectile	Charge	Temp(F)	Quantity
1	Test	Note 8	4	70	25
2	STD	Note 8	4	70	25
3	Test	Note 8	0	145	5
4	Test	Note 8	0	-50	5
5	Test	Note 8	4	145	5
6	Test	Note 8	4	-50	5

NOTES:
1. All projectiles shall be from the same lot and ± 0.05 lbs of its respective in-flight weight.
2. All rounds shall be assembled with acceptable ignition cartridges and propelling charges from the same lot.
3. All standard obturators shall be used from the same lot.
4. Group 1, 3, 4, 5 and 6 are the test groups.
5. Group 2 is the comparison group.
6. Standard obturators shall be selected from a lot of obturators that were accepted for use.
7. Rounds from group 1 and 2 shall be alternately fired in the same day.
8. To be determined based on availability.

TABLE III. Performance Test Plan.

Categories Accept/Reject	Defects		
Major:	Misfire	0	1
Major:	Obturator not discarded from the round within 20 feet from the muzzle of the tube	0	1
Major:	The average muzzle velocities of the test group 1 differ from that of the comparison group 2	(see 4.4.5.1.8)"	

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4.4.5.1.3, Delete in its entirety.

* Add new pararagraph 4.4.5.1.3 as follows:

"Flow rate. Melt flow rate shall be determined in accordance with ASTM-Method D1238."

* 4.4.5.1.5: Delete in its entirety.

Add new paragraphs 4.4.5.1.6, 4.4.5.1.7 and 4.4.5.1.8 as follows:

"4.4.5.1.6 Proving ground test. The test shall be performed at a government proving ground in accordance with the applicable USATECOM Acceptance Test Procedure.

All cartridges shall be fired from a standard, 81mm M252 mortar (hard mount) at an elevation of forty-five (45) degrees. The cartridges with the test obturators shall be separated into groups; group 1 of 25 rounds conditioned for 24 hours at +70 degrees F ± 2.5 degrees F, group 2 of 25 rounds with the standard obturator shall be conditioned at +70 degrees F ± 2.5 degrees F, group 3 of 5 rounds conditioned at +145 degrees F ± 2.5 degrees F, group 4 of 5 rounds conditioned for 24 hours at -50 degrees F ± 2.5 degrees F, group 5 of 5 rounds conditioned at +145 degrees F ± 2.5 degrees F and group 6 of 5 rounds conditioned at -50 degrees F ± 2.5 degrees F.

The individual velocities shall be recorded on the firing record. The arithmetic averages shall be calculated and recorded. Temperature of the day, weather and the time of firing shall be recorded. A smear camera shall be used to photograph all samples. The camera shall be positioned to provide photo caverage of the test rounds at 20 feet from the muzzle of the mortar tube to determine whether the obturator has come off the round. Should any round fail to meet the requirement, a photo print of these rounds, and prints of all other rounds in that particular charge zone, in that lot, shall be attached to the firing record. Film from acceptable lots may be discarded.

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4.4.5.1.7 Test validity. If for any reason ARDEC considers that the test conditions have detrimentally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

4.4.5.1.8 Statistical test. Calculate S_A for test group 1 and S_B for comparison group 2 (see Table II).

Where S_A = Standard deviation of the muzzle velocities of the twenty-five test samples.

S_B = Standard deviation of the muzzle velocities of the twenty-five comparison samples.

X_A = Average of the muzzle velocities of the twenty-five test samples.

X_B = Average of the muzzle velocities of the twenty-five comparison samples.

If $S_A^2 > S_B^2$, Calculate $\frac{S_A^2}{S_B^2} = K$ (see Notes 1 and 2).

Note 1: If $K < 2.27$ and X_A is within ± 5 fps of X_B, the test lot shall be accepted.

Note 2: If $K > 2.27$, the test shall be rejected.

Note 3: If $S_A^2 < S_B^2$ and X_A is within ± 5 fps of X_B, the test lot shall be accepted."

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6.1, Change "M853" to M853A1".

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6.4, Delete in its entirety and substitute:

"6.4 Submission of inspection equipment designs for approval. See paragraph 6.2.3 of MIL-A-48078. Submit designs as required to:

Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSTA-AR-QAA-C, Bldg 62S, Picatinny Arsenal, NJ 07806-5000."

"This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply."

* 6.5, Under Subject term (key word) listing., Add: "Flow rate" and delete: "Deflection temperature".

Add new paragraphs 6.6, 6.7 and 6.8 as follows:

"6.6 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to the following:

Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSTA-AR-QAA-C, Bldg 62S, Picatinny Arsenal, NJ 07806-5000."

"6.7 Data requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.8	Test and demonstration report	DI-T-1906

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer)."

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"6.8 Cautionary note. Polycarbonate parts are subject to chemical attack, especially by solvents. Skin contact during handling should be minimized by wearing cotton gloves. Molded obturators should be protected in bags or other containers to the greatest possible extent. In-plant storage and subsequent shipment should be in closed containers to minimize airborn contamination."

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